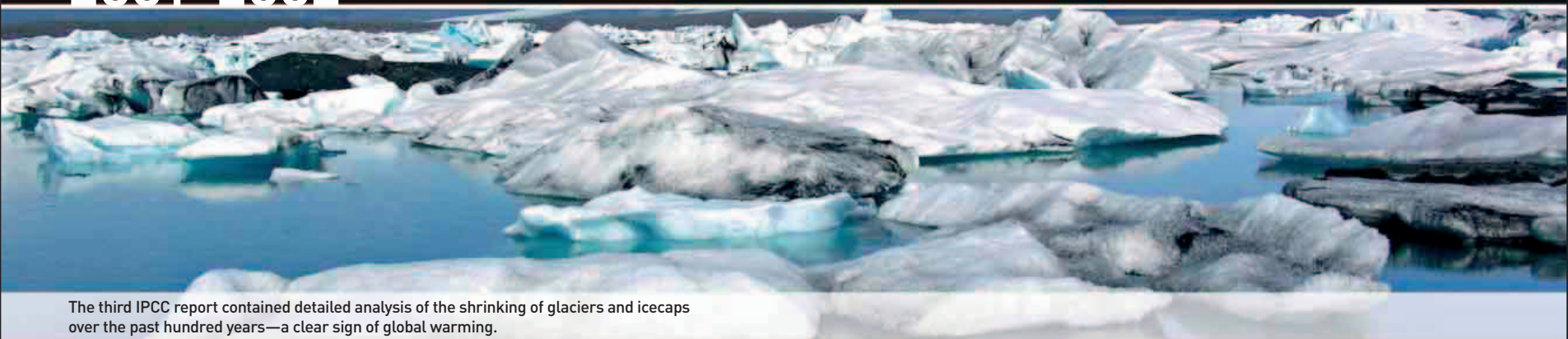




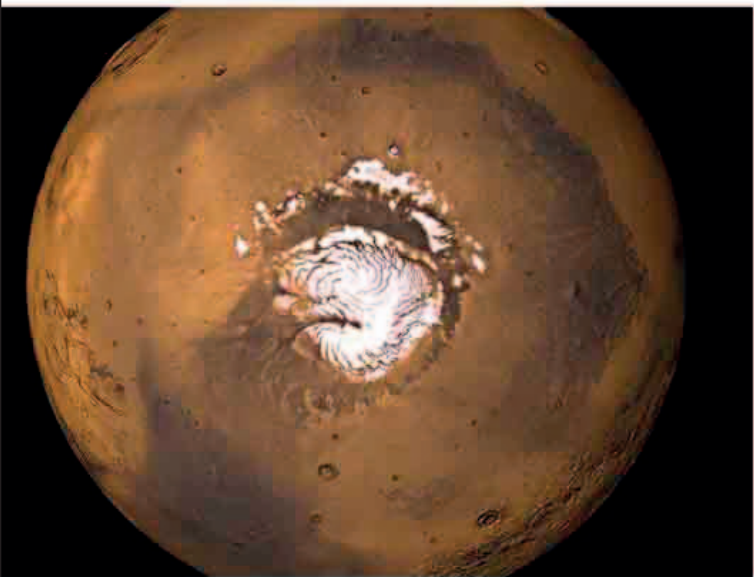
The third IPCC report contained detailed analysis of the shrinking of glaciers and icecaps over the past hundred years—a clear sign of global warming.

THE INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE

or IPCC (see 1988) published its third major report into climate change in 2001. The document supported and extended the conclusions of the organization's previous reports (see 1990, 1995) and provided more detailed projections of climate change for the decades ahead. The report included the hockey stick graph, produced by climate scientists two years earlier (see 1999).

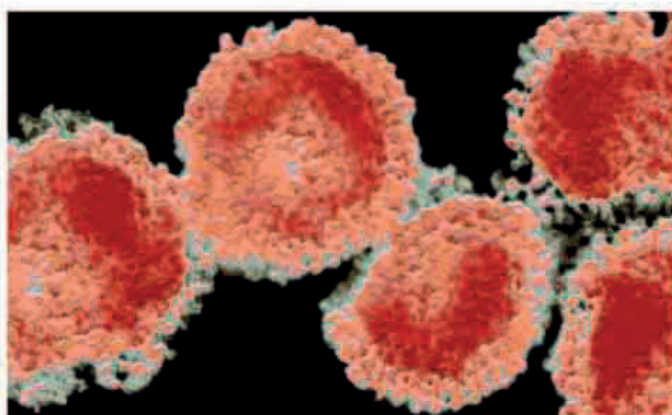
Water on Mars

This polar icecap is made of water ice covered with a layer of carbon dioxide ice. NASA's Mars Odyssey also found evidence of huge amounts of underground water.



One of the early indications of the emerging Web 2.0—in which users create content on the world wide web—was the launch of the online encyclopedia Wikipedia. Supported by the non-profit-making Wikimedia Foundation, Wikipedia can be written and edited by anyone. In 2007, it became the largest encyclopedia ever written, with more than two million articles.

In October, Apple Inc. launched its **portable digital music player**, the iPod. Similar products already existed, but sleek design and an intuitive user interface, coupled with Apple's music library and iTunes software, made the iPod a landmark consumer product.



SARS virus

This electron microscope image shows the corona virus—the virus found to be the cause of the mysterious respiratory illness known as SARS.

illness that gave people trouble breathing and, in some cases, led to their death. Treatment with antibiotics was ineffective, and researchers found none of the bacteria or viruses known to cause pneumonia. The number of cases increased rapidly and it became clear that **SARS (severe acute respiratory syndrome)** was highly infective. The disease began to spread and risked becoming a major pandemic. A combination of quarantines and the screening of airline passengers curbed the spread of the disease, and the last known case occurred in May 2004. In all, there were 8,273 reported cases, with 775 deaths—most of them in China and Hong Kong.

Researchers at the Sudbury Neutrino Observatory, Canada, detected definitive evidence of **neutrino oscillation**. Neutrinos are fundamental particles produced in nuclear reactions. There are three types, or flavors, of neutrino: muon, tau, and electron. According to the Standard Model (see 1974), neutrinos should have no mass. In the 1960s, American physicist Raymond Davis had studied solar neutrinos (see 1968), but detected one-third as many as was predicted. Davis' experiment could only detect electron neutrinos. One way to explain the deficit was that neutrinos change flavor, or oscillate, and this is what the Sudbury experiment found. Neutrino oscillation is only possible if neutrinos have mass—something the Standard Model has yet to account for.

In 2002, NASA's **Mars Odyssey space probe** detected huge amounts of water on Mars, just below the surface in the planet's Arctic region. Much of the water was locked in claylike minerals. NASA's Phoenix lander probe visited that region in 2008 and confirmed Odyssey's observations.

In Guangdong Province, China, doctors became aware of an outbreak of a pneumonialike

“ **IMAGINE A WORLD IN WHICH EVERY SINGLE PERSON... IS GIVEN FREE ACCESS TO... ALL HUMAN KNOWLEDGE. THAT'S WHAT WE'RE DOING.** ”

Jimmy Wales, founder of *Wikipedia*, 2004

January 15, 2001
Launch of the free,
online user-generated
encyclopedia Wikipedia

February 12, 2001
NASA's NEAR
Shoemaker lands
on an asteroid

March 15, 2001 The first
targeted cancer therapy,
trastuzumab (Herceptin),
completes a successful trial

April 28, 2001 American
Dennis Tito becomes the
first space tourist

May 2001 Climate Change
2001 the IPCC's Third
Assessment Report
is published

June 18, 2001
The Sudbury Neutrino
Observatory discovers
that neutrinos may
have mass

October 1, 2001
Japanese company
NTT DoCoMo launches
the first commercial
3G mobile network

October 23, 2001
Apple launches the iPod

May 28, 2002 NASA's
Odyssey probe finds
evidence of water
ice on Mars

April 15, 2002 Astronomers
detect organic compounds
in a gas cloud 26,000 light
years away

November 2002
SARS epidemic
begins in Guangdong
Province, China

December 5, 2002 Mouse
genome is published